Midterm 1

Graded

Student

Myles Sartor

Total Points

30 / 39 pts

Question 1

Problem 1

2 / 5 pts

- **– 0 pts** Correct
- **– 1 pt** (a) E and "doubles" are not independent
- ✔ **– 1 pt** (a) To show independence of two events A and E, need to show $P(E \cap A) = P(E)P(A)$
- ✔ **– 0 pts** (a) mutually exclusive doesn't mean independence. Saying (4,4) is contained in both events A and E doesn't imply "doubles" and E are independent
- ✔ **– 1 pt** (b) $P(E) = \frac{3}{25}$
- **– 1 pt** (b) $P(E \cap \text{"at least one 3"}) = \frac{2}{25}$
- ✔ **– 1 pt** $P(\text{"at least one 3"}|E) = \frac{2}{25}/\frac{3}{25} = \frac{2}{3}$

Question 2

Problem 2

8 / 8 pts

- ✔ **– 0 pts** Correct
- **– 2 pts** (a) Numerator
- **– 2 pts** (a) Denominator
- **– 2 pts** (b) Numerator
- **– 1 pt** (b) Denominator: total number of possibilities
- **– 1 pt** (b) Denominator: |C'| that needs to be subtracted
- **– 1 pt** mistake (noted with 'x')
- **– 8 pts** Incorrect

Question 3

Problem 3

5.5 / 6.5 pts

- **– 0 pts** Correct
- **– 2 pts** part a: incorrect answer and did not show any work
- **– 1.5 pts** part a: major error in understanding problem
- **– 1 pt** part a: correct answer but no set up or work done
- **– 1 pt** part a:used the incorrect lower bound
- **– 1 pt** part a: did not set equal to 1
- **– 1 pt** part a: error in integration (did not integrate the bounds correctly or said $\int \cos(t) = -\sin(t)$)
- **– 0.5 pts** part a: correct up until finding which k solves $\sin(k) = 1$
- **– 0.5 pts** part a: forgot a factor of 1/2 in the problem
- **– 0.5 pts** part a: correct up until some minor algebraic error
- **– 2.5 pts** part b: no answer given
- **– 2 pts** part b: major error in understanding what to do for cdf
- **– 1.5 pts** Part b: incorrect answer and did not do the integration set up: $F(t) = \int_{-\infty}^{t} f(t)dt$

  possibly confused pdf with cdf
- ✔ **– 1 pt** part b: did not integrate $\int_{-\frac{\pi}{2}}^{a} \frac{\cos(t)}{2}$ correctly but at least set the problem up correctly
- **– 0.5 pts** Part b: forgot the factor of $\frac{1}{2}$
- **– 0.5 pts** part b:minor algebraic error after correct set up
- **– 0.5 pts** part b: error in writing the piecewise function, did not include where the cdf takes certain values (when is it 0, when is it 1, etc.)
- **– 2 pts** part c missing
- **– 1.5 pts** part c: major error in finding the probability
- **– 1 pt** Part c: error in integrating the pdf
- **– 1 pt** part c: said $F(\frac{\pi}{6}) = 0$ or $F(\frac{\pi}{2}) = 0$
- **– 0.5 pts** Part c: error in integrating the pdf but still got correct final answer
- **– 0.5 pts** part c: forgot a factor of 1/2
- **– 0.5 pts** part c: did not evaluate $\sin(\frac{\pi}{2})$ and $\sin(\frac{\pi}{6})$ at all or correctly
- **– 0.25 pts** part c: did not evaluate $\sin(\frac{\pi}{6})$ or $\sin(\frac{\pi}{2})$ correctly (or at all)
- **– 0.25 pts** part c: did not simplify all the way
- **– 0.25 pts** part c: bounds evaluated incorrectly
- **– 6.5 pts** no answer

Question 4

Problem 4

6 / 6 pts

✔ **− 0 pts** Correct

− 1 pt (a) incorrect pmf X values (should be discrete)

− 1 pt (a) incorrect pmf probability values

− 0.5 pts (b) $P_X(2) = F_X(2^+) - F_X(2^-)$

− 0.5 pts (b) $P_X(2) = \frac{3}{36} = \frac{1}{12}$

− 0.5 pts (c) $P(1 \leq X \leq 4) = F(4) - F(1^-)$: wrong $F(1^-)$

− 0.5 pts (c) $P(1 \leq X \leq 4) = F(4) - F(1^-)$: wrong $F(4)$

− 1 pt (c) $P(1 \leq X \leq 4) = \frac{16}{36} - 0 = \frac{4}{9}$

− 0.5 pts (a) algebraic mistake

− 3 pts pmf missing

− 1.5 pts should use pmf values but used cdf values

− 1 pt incomplete pmf

Question 5

Problem 5

1 / 5 pts

− 0 pts Correct

✔ **− 1 pt** (a) formula $P(F) = P(F|E)P(E) + P(F|E')P(E')$

✔ **− 0.5 pts** (a) $P(F|E)$

✔ **− 0.5 pts** (a) $P(E)$

✔ **− 0.5 pts** (a) $P(F|E')$

✔ **− 0.5 pts** (a) $P(E')$

− 1 pt (b) formula

✔ **− 0.5 pts** (b) numerator

✔ **− 0.5 pts** (b) denominator

− 5 pts Incorrect

− 1 pt (a) answer

Question 6

Problem 6

4 / 4 pts

✔ **− 0 pts** Correct

− 1.5 pts part a: some major error in finding probability

− 1 pt part a: confusion of exactly one with at least one

− 0.5 pts part a: set up correct but error in calculating probability

− 0.25 pts part a: algebraic error

− 2 pts part b: missing

− 1.5 pts part b: major error in finding probability

− 1 pt part b: did not use conditional probability rules

− 1 pt part b: treated the events A and D as independent events by implying $P(A \cap D) = P(A)P(D)$

− 1 pt part b: did not divide by $P(D)$

− 1 pt part b: error in numerator

− 1 pt part b: problem not completed

− 0.25 pts part b: did not simplify

− 4 pts problem missing

− 0 pts Click here to replace this description.

Question 7

Problem 7

2 / 2 pts

✔ **− 0 pts** Correct

− 1 pt a incorrect

− 1 pt b incorrect

Question 8

Problem 8

1 / 1.5 pts

− 0 pts Correct

− 0.5 pts (a) True

− 0.5 pts (b) False

✔ **− 0.5 pts** (c) False

Question 9

Problem 9

0.5 / 1 pt

− 0 pts Correct

✔ **− 0.5 pts** (a) False: pariwise independence doesn't mean $P(A \cap B \cap C) = P(A)P(B)P(C)$

− 0.5 pts (b) True: $P(B'|C) = P(B')$ because B' and C are independent as B and C are independent

Midterm 1-B

Name: Myles Sartor Student ID: 119017708

Problem 1. [Total 5 points]

You roll two five-sided dice. The sides of each die are numbered from 1 to 5. The dice are "fair"" (all sides are equally likely), and the two die rolls are independent. Event E is "the total is 8."

(a) Is event E independent of getting "doubles" (i.e., both dice resulting in the same number)? Show your work. [2 points]

(b) Given that the total was 8, what is the probability that at least one of the dice resulted in a 3?[3 points]

a) No, event E is not independent of getting doubles, as one possibility within the event results in an outcome of (4,4), which is an outcome that has a double adding to eight.

b) $\frac{2}{25}$, as only outcomes (3,5), and (5,3) result in outcomes with a 3 in at least one of the dice.

↳The other was (4,4)

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
(3,1)	(3,2)	(3,3)	(3,4)	~~(3,5)~~
(4,1)	(4,2)	(4,3)	~~(4,4)~~	(4,5)
(5,1)	(5,2)	~~(5,3)~~	(5,4)	(5,5)

Midterm 1-B

Name: Myles Sartor Student ID: 119617708

Problem 2. [Total 8 points]

3 Three cards are randomly selected from an ordinary deck of 52 cards. Let A be the event that all cards are Queens. Let B be the event that Queen of diamonds is chosen and C be the event at least one ~~Ace~~ Queen is chosen. Find

(a) $P(A|B)$ [4 points]

(b) $P(A|C)$ [4 points]

A = all queens (4)

B = Queen of diamonds (1)

$\binom{52}{3} = S$

$\binom{4}{3}$

a) $P(A|B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{\binom{3}{2}}{\binom{51}{2}}$

$A \subseteq C$

↳ A is a subset of C

b) $P(A|C) = \dfrac{P(A \cap C)}{P(C)} = \dfrac{\binom{4}{3}}{\binom{52}{3} - \binom{48}{3}}$

Complement: $\binom{48}{3}$

Midterm 1-B

Name: Myles Sartor   Student ID: 119617708

Problem 3. [6 points] Random variable T is distributed with the following pdf:

$\frac{t\sin t}{2}$ $\frac{1}{2} t\sin t$

$$f(t) = \begin{cases} \frac{cos(t)}{2} & -\frac{\pi}{2} < t < k \\ 0 & \text{otherwise} \end{cases}$$

$\sin t\Big|_{-\pi/2}^{K} \rightarrow \sin K - \sin(-\frac{\pi}{2}) = 2$

$\sin K + 1 = 2$

$\sin K = 1$

$K = \frac{\pi}{2}$

(a) What is the value of the constant k?[2 point]

(b) What is the corresponding CDF?[2.5 point]

(c) Calculate the probability $P(\frac{\pi}{6} \leq T \leq \frac{\pi}{2})$? [2 point]

$\int_{-\pi/2}^{K} \cos t = 2$

$\int_{-\pi/2}^{K} \frac{\cos t}{2} = 1$

a) $\int_{-\pi/2}^{K} \frac{\cos(t)}{2}$ $K = \frac{\pi}{2}$

$\int_{-\pi/2}^{K} \frac{\cos t}{2} = \frac{t \sin t}{2}\Big|_{-\pi/2}^{K}$

$= \frac{K \sin K}{2} - \frac{-\frac{\pi}{2}\sin(-\frac{\pi}{2})}{2}$

b) $$F(T) = \begin{cases} 0 & t < 0 \\ \frac{1}{2} t \sin t & -\frac{\pi}{2} < t < \frac{\pi}{2} \\ 1 & t > 1 \end{cases}$$

2

$\sin(\pi/6) = \frac{1}{2}$

c) $P(\frac{\pi}{6} \leq T \leq \frac{\pi}{2}) = F(\frac{\pi}{2}) - F(\frac{\pi}{6}^-)$

$\frac{1}{2}(\frac{\pi}{2})\sin(\frac{\pi}{2}) - \frac{1}{2}(\frac{\pi}{6})\sin(\frac{\pi}{6})$

?

$= 1 - \frac{1}{2} = \frac{1}{2}$

Name: Myles Sartor Student ID: 114017708

Problem 4. Assume that CDF of the random variable X is given as follows.

$$F_X(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{36} & 1 \le x < 2 \\ \frac{4}{36} & 2 \le x < 3 \\ \frac{9}{36} & 3 \le x < 4 \\ \frac{16}{36} & 4 \le x < 5 \\ \frac{25}{36} & 5 \le x < 6 \\ 1 & x \ge 6 \end{cases}$$

$F(x) - F(x^-)$

(a) Find pmf of X.[3 points]

(b) Using CDF find $P_X(2)$[1 point]

(c) $P(1 \le X \le 4)$ [2 points]

$P(x=0) = 0$

$P(x=1) = \frac{1}{36} - 0$

$P(x=2) = \frac{4}{36} - \frac{1}{36}$

$P(x=3) = \frac{9}{36} - \frac{4}{36}$

$P(x=4) = \frac{16}{36} - \frac{9}{36}$

$P(x=5) = \frac{25}{36} - \frac{16}{36}$

$P(x=6) = 1 - \frac{25}{36}$

a)

X	1	2	3	4	5	6
$P_X(x)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

b) $P_X(2) = F(2) - F(2^-)$ $x<2$

$\hookrightarrow \frac{4}{36} - \frac{1}{36} = \frac{3}{36}$

c) $P(1 \le x \le 4) = F(4) - F(1^-)$ $x<1$

$\hookrightarrow \frac{16}{36} - 0 = \frac{16}{36}$

Midterm 1-B

Name: Myles Sartor Student ID: 119017708

5 3 2

Problem 5. Box A contains five red and three blue balls, and box B contains two red and three blue balls. A ball is drawn at random from box A and transferred to box B.

3 $5+3+2+3=13$

(a) Compute the probability of drawing a blue ball from box B. [3 points]

(b) If a blue ball was drawn from box B, what is the probability that a red ball had been drawn from box A? [2 points]

$3+3=6$

6

$5+2=7$

A = Ball drawn from box A

a) $\binom{18}{6}$

B = Ball transferred to box B

$P(A|B)$

$P(B|A)$

b) $P(A|B) = \dfrac{P(A\cap B)}{P(B)} = \dfrac{\binom{8}{5}\binom{5}{3}}{\binom{18}{6}}$

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Problem 6. Let A, B and C be mutually independent and $P(A) = 1/2$, $P(B) = 1/3$, $P(C) = 1/4$. Let D be the event that exactly one of events A, B and C occurs.

(a) Compute $P(D)$. [2 points]

(b) Compute $P(A|D)$. [2 points]

a) $P(D) = P(A \cap B' \cap C') + P(A' \cap B \cap C') + P(A' \cap B' \cap C)$

$\hookrightarrow \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{1}{4}$

$= \frac{6}{24} + \frac{3}{24} + \frac{2}{24} = \frac{11}{24}$

b) $P(A|D) = \frac{P(A \cap D)}{P(D)} = \frac{\frac{6}{24}}{\frac{11}{24}} = \frac{6}{11}$

Midterm 1-B

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Problem 7. Let A, B, and C be three events. Interpret the following events with set theory notations (intersections, complement, unions, etc)

a. [1 point] only A occurs; $P(A \cap B' \cap C')$

b. [1 point] both B and C, but A does not occur; $P(A' \cap B \cap C)$

Problem 8. Let A,B, and C be disjoint subsets of the sample space. For each one of the following statements, determine whether it is true or false.

a. [0.5 point] $P(A) + P(B) \leq 1$ True

b. [0.5 point] $P(C') + P(B) \leq 1$ False

c. [0.5 point] $P(A) + P(B) + P(C) = 1$ True

$1 - P(C) + P(B) = 1$

$-P(C) + P(B) = 0 \rightarrow P(B) = P(C) \rightarrow$ Not always true

Problem 9. Let A,B, and C are pairwise independent subsets of the sample space. For each one of the following statements, determine whether it is true or false.

a. [0.5 point] $P(A|B \cap C) = P(A)$ True

b. [0.5 point] $P(B'|C) = P(B')$ True

$$P(A|B \cap C) = \frac{P(A \cap (B \cap C))}{P(B \cap C)}$$

$$P(B'|C) = \frac{P(B' \cap C)}{P(C)}$$

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